

## UNIT-1

Introduction – Network hardware – software – reference models – OSI and TCP/IP models – Example Networks: Internet, ATM, Ethernet and Wireless LANs – Physical Layers – Theoretical basis for data communication - Guided Transmission Media.

### **1.1 INTRODUCTION – COMPUTER NETWORK.**

- ❖ A collection of autonomous computers interconnected by a single technology.
- ❖ Two computers are said to be interconnected if they are able to exchange information.
- ❖ The collection need not be via a copper wire; fiber optics, microwaves, infrared and communication satellites can be used.

#### **Internet And World Wide Web**

- ❖ The internet is a massive network of networks, It connects millions of computers together globally, forming a network in which any computer can communicate with any other computer as long as they are both connected to the internet. Information that travels over the internet does so via a variety of languages known as protocols.
- ❖ The world wide web or simply web is a way of accessing information over the medium of the internet.

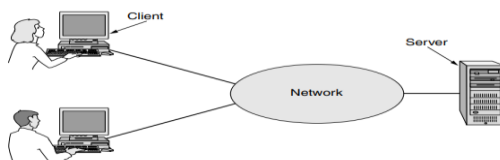
#### **Internet And Distributed System**

- ❖ The Internet is not a single network, but a network of networks.
- ❖ In a distributed system, a collection of independent computers appears to its users as a single coherent system.
- ❖ A well known example of a distributed system is the World Wide Web, in which everything looks like a document.

#### ***Uses of Computer Networks***

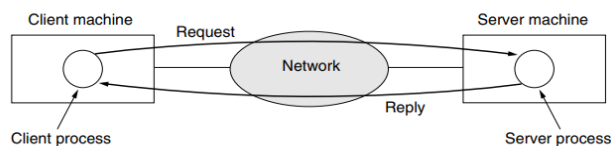
**Business Applications:** Most companies have a substantial number of computers. For example, a company may have a computer for each worker and use them to design products, write brochures, and do the payroll.

- The client and server machines are connected by a network, as illustrated in Fig. 1-1



**Figure 1-1.** A network with two clients and one server.

- The detailed client server model illustrated in fig. 1-2.



**Figure 1-2.** The client-server model involves requests and replies.

- The company employee can share their information through e-mail.
- The company employee can share their ideas through video conferencing.
- Many companies are doing business electronically with other companies, especially suppliers and customers.
- Doing business with customers over the Internet. Many companies provide catalogs of their goods and services online and take orders online. It is called e-commerce.

**Home Applications:** The main usage of networks in home is internet. Some popular uses of the Internet for home users are:

**Access to remote information** by the surfing the world wide web for information or just for fun.

**Person-person communication** to distinguish it from the client- server model in this form, individuals who form a loose group can communicate with others in the group.

**Interactive entertainment** where any movie or television program made in any country can be displayed on the screen instantly.

**Electronic commerce** where people do home shopping, pay their bills, manage their bank accounts and handle their investments electronically.

**Mobile Users:** Mobile computers, such as notebook computers and personal digital assistants (PDAs) are connected to the office or home even when away from home.

- People on the road use their portable electronic equipment to send and receive telephone calls, faxes, and electronic mail, surf the Web, access remote files, and log on to remote machines.
- People do this from anywhere on land, sea, or air.
- SMS, M-Commerce, GPS.

**Social issues:** newsgroups people can exchange message with like-minded individuals.

- Employee rights versus employer rights. Many people read and write e-mail at work.
- The government does not have a monopoly on threatening people's privacy. The private sector does its bit too.
- Computer networks offer the potential for sending anonymous messages.
- A lot of these problems could be solved if the computer industry took computer security seriously.

## **1.2 NETWORK HARDWARE**

- It depends on transmission technology and scale.

### 1. Transmission Technology

There are two types of transmission technology:

1. Broadcast links.
2. Point-to-point links.

#### 1.1 Broadcast link

- Broadcast networks have a single communication channel that is shared by all the machines on the network.
- Short messages, called packets sent by any machine are received by all the others. An address field within the packet specifies the intended recipient.
- On receiving a packet, a machine checks the address field.
- If the packet is intended for the receiving machine, that machine processes the packet; if the packet is intended for some other machine, it is just ignored.
- Broadcast systems also allow the possibility of addressing a packet to all destinations by using a special code in the address field.
- Some broadcast systems support transmission to a subset of the machines, called as multicasting.

#### 1.2 Point-to-point link

- Point-to-point networks consist of many connections between individual pairs of machines.

- Point-to-point transmission with one sender and one receiver is sometimes called unicasting.

## 2. Scale

- An alternative criterion for classifying networks is their scale.
- The table classifies networks according to their scale.

| Interprocessor distance | Processors located in same | Example                   |
|-------------------------|----------------------------|---------------------------|
| 1 m                     | Square meter               | Personal area network     |
| 10 m                    | Room                       | Local area network        |
| 100 m                   | Building                   |                           |
| 1 km                    | Campus                     |                           |
| 10 km                   | City                       | Metropolitan area network |
| 100 km                  | Country                    | Wide area network         |
| 1000 km                 | Continent                  |                           |
| 10,000 km               | Planet                     | The Internet              |

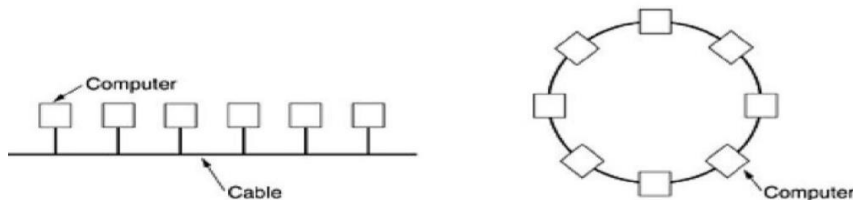
Fig. classification of interconnected processors by scale.

### Personal Area Networks (PANs)

- It communicates over the range of person.
- Its coverage area of the network is within a meter. Ex. Bluetooth.
- A wireless network connecting a computer with its mouse, keyboard and printer is a personal area network.
- PANs are also used in mobile headsets, car connections and medical devices (pacemaker, insulin pump) and can be built using technologies like RFID.

### Local Area Networks (LANs)

- Local area networks, called LANs, are privately-owned networks within a single building or campus of up to a few kilometres in size.
- They are widely used to connect personal computers and workstations in company offices and factories to share resources (e.g., printers) and exchange information.
- LANs are distinguished by three characteristics: (1) their size(distance), (2) their transmission technology(broadcast or point-to -point) and (3) topology(bus or ring).
- LANs are restricted in size. LANs use a transmission technology consisting of a cable to which all the machines are attached.
- Various topologies are used for broadcast LANs.

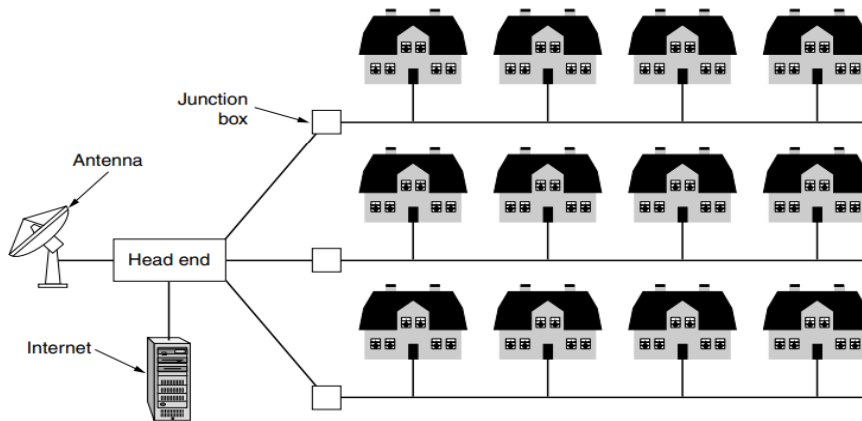


- In a bus network, at any instant at most one machine is the master and is allowed to transmit. All other machines are required to refrain from sending. Eg: IEEE 802.3(Ethernet).
- In the ring system each bit propagates around on its own, not waiting for the rest of the packet to which it belongs.

- The cable to which the nodes connect is called a “ backbone ”. If the backbone is broken, the entire segment fails.
- Data travels from node to node, with each node along the way handling every packets.

### Metropolitan area networks (MAN)

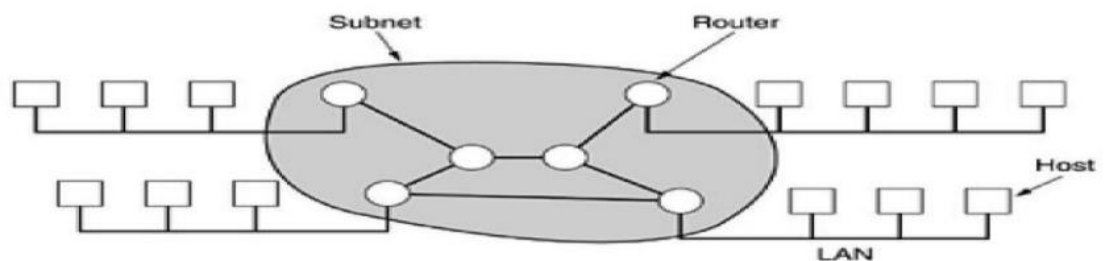
- A metropolitan area network, or MAN, covers a city like cable television network available in many cities.
- A metropolitan area network (MAN) is similar to a local area network (LAN) but spans an entire city or campus.
- MANs are formed by connecting multiple LANs. Thus, MANs are larger than LANs but smaller than wide area networks (WAN).
- A metropolitan area networks as shown in fig.



**Fig : A metropolitan area network based on cable TV.**

### Wide Area Networks (WANs)

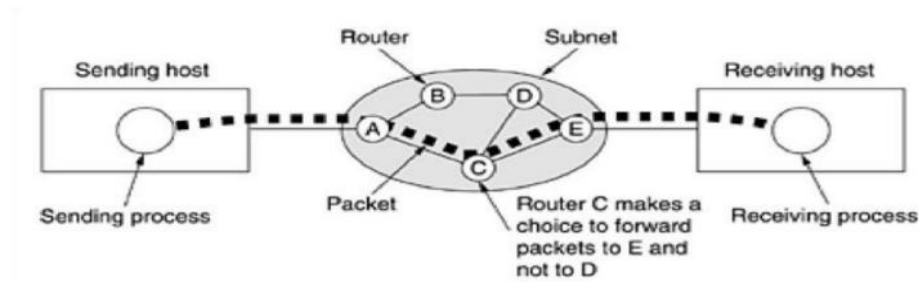
- A wide area network, or WAN, covers large geographical area, often a country or continent.
- It contains a collection of machines intended for running user programs.
- The hosts are connected by a communication subnet.
- The **subnet** is defined as the collection of routers and communication lines that moved packets from the source host to the destination host.
- The hosts are owned by the customers, whereas the communication subnet is owned and operated by a telephone company or Internet service provider.
- The job of the subnet is to carry messages from host to host.



**Fig. Relation between hosts on LANs and the subnet.**

- In most wide area networks, the subnet consists of two distinct components: transmission lines and switching elements.
- Transmission lines move bits between machines. They can be made of copper wire, optical fiber, or even radio links.

- Switching elements are specialized computers that connect three or more transmission lines.
- When data arrive on an incoming line, the switching element must choose an outgoing line on which to forward them. These switching computers are called router.
- A stream of packets from sender to receiver is represented in fig.



- In this fig. all the packets follow the route ACE , rather than ABDE or ACDE. ACE is the best route.

### Internetworks

- The world collection of computer (networks) is called internetworks or internet.
- A common form of internet is a collection of LANs connected by a WAN.
- Gateway: it is used to connection between two or more networks

### Wireless Networks:

A Wireless network is a computer network that uses wireless data connection between nodes.

Wireless networks can be divided into three main categories:

- **System interconnection-** It is interconnecting the components of a computer using short-range radio.
- Some short-range wireless networks called Bluetooth are used to connect these components without wires.
- **Wireless LANs** or WLAN is a network that allows devices to connect and communicate wirelessly.
- **Wireless WANs.**

### Home Area Networks

Every device in the home is capable of communicating with every other device, and all of them will be accessible over the Internet. Some categories are:

- ✓ Computers (desktop PC, notebook PC, PDA, shared peripherals).
- ✓ Entertainment (TV, DVD, VCR, MP3, camera, stereo).
- ✓ Telecommunications (telephone, mobile telephone, intercom).
- ✓ Appliances (microwave, refrigerator, clock, lights).
- ✓ Telemetry (utility meter, smoke/burglar alarm).

## 1.3 NETWORK SOFTWARE

### Protocol Hierarchies

- Networks are organized as a stack of layers or levels, each one built upon the other.

- The number of layers, the name of each layer, the contents of each layer, and the function of each layer differ from network to network.
- The purpose of each layer is to offer certain services to the higher layers.

A five-layer network is:

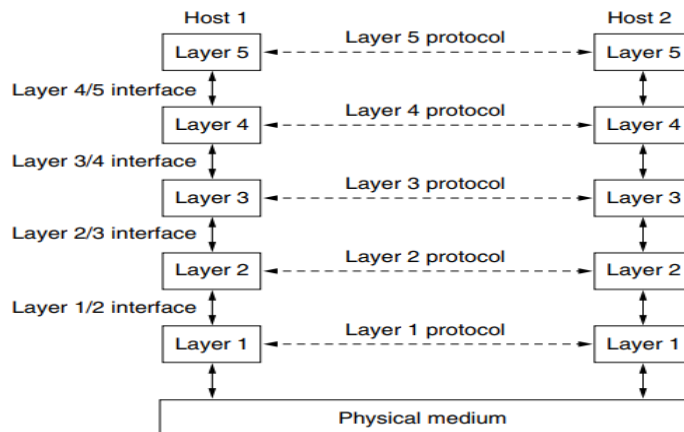


Fig. layers, protocols and interface.

- The entities comprising the corresponding layers on different machines are called peers. The peers may be processes, hardware devices, or even human beings.
- No data are directly transferred from layer  $n$  on one machine to layer  $n$  on another machine. Each layer passes data and control information to the layer immediately below it, until the lowest layer is reached.
- Below layer 1 is the physical medium through which actual communication occurs.
- A set of layers and protocols is called a network architecture. A list of protocols used by a certain system, one protocol per layer, is called a protocol stack.

### Design issues

- Every layer needs a mechanism for identifying senders and receivers. Since a network normally has many computers.
- The rules for data transfer. In some systems, data only travel in one direction; in others, data can go both ways. And also it specifies the priorities of data transfer either normal or urgent data transfer.
- Error control is an important issue because physical communication circuits are not perfect. Error correction and detection mechanism to handle error during transfer.
- Not all communication channels preserve the order of messages sent on them. It is possible to loss of sequencing.
- An issue that occurs at every level is how to keep a fast sender from swapping a slow receiver with data.
- The inability of all processes to accept arbitrarily long messages. This property leads to mechanism for disassembling, transmitting and then reassembling messages.
- When there are multiple paths between source and destination a route must be chosen.

### Connection-Oriented and Connectionless Services

- Layers offer two different types of service to the layers above them: connection-oriented and connectionless.
- **Connection-oriented service** first establishes a connection, uses the connection, and then releases the connection. eg. Telephone system.
- The sender pushes objects (bits) in at one end, and the receiver takes them put at the other end.

- A reliable service is implemented through acknowledging concept.
- The acknowledge concept introduces overhead and time delay. It is not possible to lose the data.
- Example: TCP
- **Connectionless service** is modelled after the postal system. Each message carries the full destination address, and each one is routed through the system independent of all the others.
- Normally, when two messages are sent to the same destination, the first one sent will be the first one to arrive. Sometimes the order may be vary.
- A non-reliable service is implemented with faster file transfer. It is possible to lose the data.
- The internet protocol (IP) and User Datagram Protocol(UDP) are connectionless protocols.

|                     |                         |                      |
|---------------------|-------------------------|----------------------|
| Connection-oriented | <b>Service</b>          | <b>Example</b>       |
|                     | Reliable message stream | Sequence of pages    |
|                     | Reliable byte stream    | Movie download       |
| Connection-less     | Unreliable connection   | Voice over IP        |
|                     | Unreliable datagram     | Electronic junk mail |
|                     | Acknowledged datagram   | Text messaging       |
|                     | Request-reply           | Database query       |

Fig. six different types of service.

### Service Primitives

A service is specified by a set of primitives available to a user process to access the service. The primitives are normally system calls of an OS.

| Primitive  | Meaning                                    |
|------------|--------------------------------------------|
| LISTEN     | Block waiting for an incoming connection   |
| CONNECT    | Establish a connection with a waiting peer |
| ACCEPT     | Accept an incoming connection from a peer  |
| RECEIVE    | Block waiting for an incoming message      |
| SEND       | Send a message to the peer                 |
| DISCONNECT | Terminate a connection                     |

Fig. Six service primitives that provide a simple connection-oriented service.

- The server executes LISTEN to indicate that it is prepared to accept incoming connections.
- Next, the client process executes CONNECT to establish a connection with the server.
- The next step is for the server to executes RECEIVE to prepare to accept the first request.
- Then the client executes SEND to transmit its request followed by the execution of RECEIVE to get the reply.
- If the data transferred successfully then it can use DISCONNECT to terminate the connection.

Briefly summarizes how client-server communication might work over a connection -oriented network.

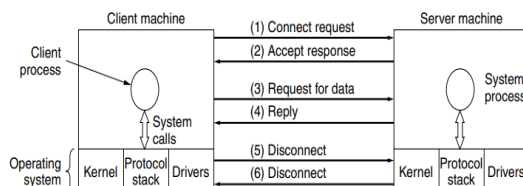


fig. A simple client-server interaction using acknowledged datagrams.

## Relationship of Services and Protocols

- A service is a set of primitives that a layer provides to the layer above it.
- The service defines what operations the layer is prepared to perform on behalf of its users, but it says nothing at all about how these operations are implemented.
- A service relates as an interface between two layers, with the lower layer being the service provider and the upper layer being the service user.
- A protocol is a set of rules governing the format and meaning of the packets, or messages that are exchanged by the peer entities within a layer.
- Services relate to the interfaces between layers whereas protocols relate to the packets sent between peer entities on different machines.

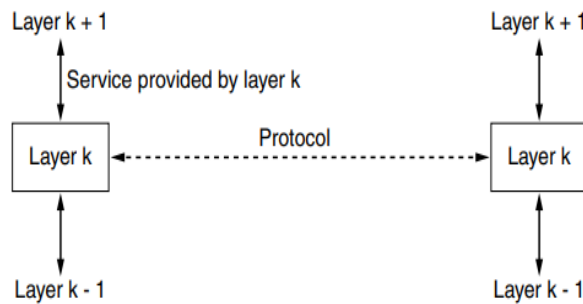


Fig. the relationship between a service and a protocol.

## 1.4 REFERENCE MODEL

- OCI Reference Model
- TCP/IP Reference Model

### 1.4.1 OCI Reference model

- This model is based on a proposal developed by the International Standards Organization (ISO).
- It is called the ISO OSI (Open Systems Interconnection) Reference Model because it deals with connecting open systems that is, systems that are open for communication with other systems.

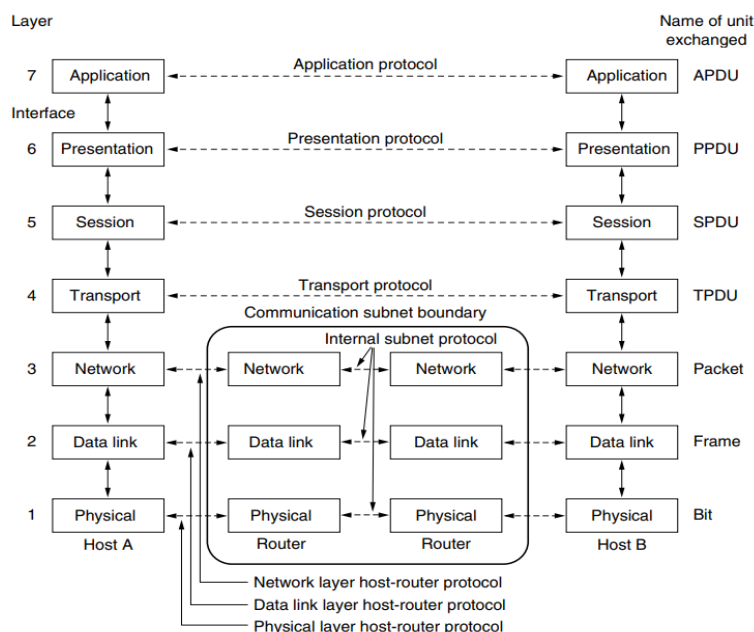


Fig . the OCI reference model.



The OSI model has seven layers. The principles that were applied to arrive at the seven layers can be briefly summarized as follows:

1. A layer should be created where a different abstraction is needed.
2. Each layer should perform a well-defined function.
3. The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
4. The layer boundaries should be chosen to minimize the information flow across the interfaces.
5. The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.

### **The Physical Layer**

- The physical layer is concerned with transmitting raw bits over a communication channel.
- The design issues have to do with making sure that when one side sends a 1 bit it is received by the other side as a 1 bit, not as a 0 bit.
- These design issues largely deal with mechanical, electrical, and timing interfaces, as well as the physical transmission medium, which lies below the physical layer.

The components of the physical layer are:

- Cabling system components
- Adapters that connect media to physical interfaces
- Hub, repeater, and patch panel specifications
- Network Interface Card (NIC)

### **The Data Link Layer**

- The main task of the data link layer is to transform a raw transmission facility into a line that appears free of undetected transmission errors to the network layer.
- It performs this task by the sender breaking up the input data into data frames and transmit the frames sequentially.
- If the service is reliable, the receiver confirms correct receipt of each frame by sending back an acknowledgement frame.

This layer provides the following functions:

- Allows a device to access the network to send and receive messages
- Offers a physical address so a device's data can be sent on the network
- Works with a device's networking software when sending and receiving messages.

Provides error-detection capability The common networking components are :

- Network interface cards
- Ethernet and Token Ring switches
- Bridges

### **The Network Layer**

- The network layer controls the operation of the subnet.

- A key design issue is determining how packets are routed from source to destination. The quality of service provided is also a network layer issue.
- When a packet has to travel from one network to another to get to its destination, many problems can arise.
  - The addressing used by the second network may be different from that used by the first one.
  - The second one may not accept the packet at all because it is too large. The protocols may differ, and so on.
  - It is up to the network layer to overcome all these problems to allow heterogeneous networks to be interconnected.
- Network layer overcomes all these problems to allow heterogeneous networks to be interconnected.

### **The Transport Layer**

- The basic function is to accept data from above, split it up into smaller units, pass these to the network layer, and ensure that the pieces all arrive correctly at the other end.
- The transport layer also determines what type of service to provide to the session layer, and, to the users of the network.

Functions:

- Segmentation of data for network transport
- Control of data flow to prevent memory overruns
- Transmission-error detection The most common transport layer protocols are the connection-oriented TCP Transmission Control Protocol (TCP) and the connectionless UDP User Datagram Protocol (UDP).

### **The Session Layer**

- The session layer allows users on different machines to establish sessions between them.
- Sessions offer various services including,
  - dialog control
  - token management
  - synchronization.

### **The Presentation Layer**

- It is concerned with the syntax and semantics of the information transmitted.
- In order to make it possible for computers with different internal data representations to communicate, the data structures to be exchanged can be defined in an abstract way, along with a standard encoding to be used “on the wire.”
- The presentation layer manages abstract data structures and allows higher-level data structures (e.g., banking records), to be defined and exchanged.

### **The Application Layer**

- The application layer contains a variety of protocols that are commonly needed by users.
- One widely used application protocol is HTTP (HyperText Transfer Protocol), which is the basis for the World Wide Web.
- When a browser wants a Web page, it sends the name of the page it wants to the server hosting the page using HTTP.
- The server then sends the page back. Other application protocols are used for file transfer, electronic mail, and network news.

### 1.4.1. TCP/IP Reference Model

- The ARPANET was a research network sponsored by U.S. Department of Defense with a major goal to connect multiple networks.
- the ability to connect multiple networks in a seamless way was one of the major design goals. This architecture later became known as the TCP/IP Reference Model
- This architecture was later known as the TCP/IP Reference Model, after its two primary protocols.

#### Internet layer

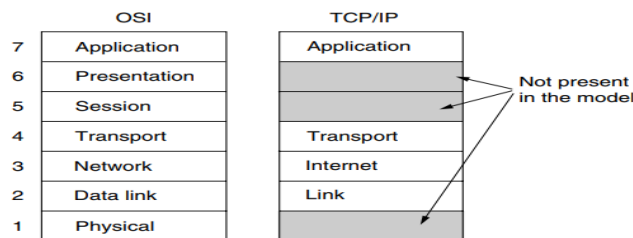
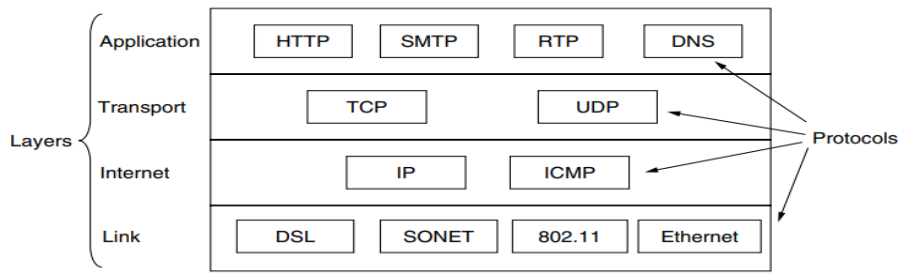


Figure 1-21. The TCP/IP reference model.

- This layer permits hosts to inject packets into any network and have them travel independently to the destination.
- They may even arrive in a different order than they were sent, in which higher layers rearrange them.
- It defines an official packet format and protocol called IP (Internet Protocol).
- The job of the internet layer is to deliver IP packets where they are supposed to go. Packet routing is clearly the major issue.
- The Internet layer is responsible for addressing, packaging, and routing functions. The core protocols of the Internet layer are IP, ARP, ICMP, and IGMP.
- The Internet Protocol (IP) is a routable protocol responsible for IP addressing, routing, and the fragmentation and reassembly of packets.
- The Address Resolution Protocol (ARP) is responsible for the resolution of the Internet layer address to the Network Interface layer address such as a hardware address.
- The Internet Control Message Protocol (ICMP) is responsible for providing diagnostic functions and reporting errors due to the unsuccessful delivery of IP packets.
- The Internet Group Management Protocol (IGMP) is responsible for the management of IP multicast groups.

#### The Transport Layer

- The layer above the internet layer is called the transport layer. It is designed to allow peer entities on the source and destination hosts to carry on a conversation.
- TCP (Transmission Control Protocol), is a reliable connection-oriented protocol that allows a byte stream originating on one machine to be delivered without error on any other machine in the internet.
- It fragments the incoming byte stream into discrete messages and passes each one on to the internet layer.
- At the destination, the receiving TCP process reassembles the received messages into the output stream.
- UDP (User Datagram Protocol), is an unreliable, connectionless protocol for applications that do not want TCP's sequencing or flow control and wish to provide their own. The relation of IP, TCP, and UDP is:



**Figure 1-22.** The TCP/IP model with some protocols we will study.

## The Application Layer

- The TCP/IP model does not have session or presentation layers.
- On top of the transport layer is the application layer. It contains all the higher-level protocols. The early ones included virtual terminal (TELNET), file transfer (FTP), and electronic mail (SMTP). Many other protocols have been added to these over the years. Some important ones that we will study, shown in Fig.
- include the Domain Name System (DNS), for mapping host names onto their network addresses, HTTP, the protocol for fetching pages on the World Wide Web, and RTP, the protocol for delivering real-time media such as voice or movies.

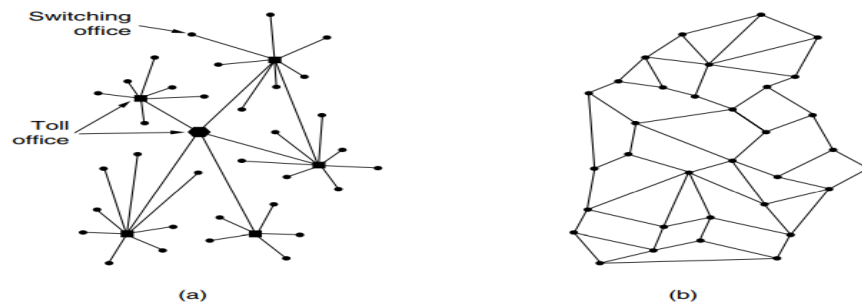
## 1.5 EXAMPLE NETWORKS

### Internet

- The number of networks, machines, and users connected to the ARPANET grew rapidly after TCP/IP became the only official protocol on January 1, 1983
- When NSFNET and the ARPANET were interconnected, the growth became exponential. Many regional networks joined up, and connections were made to networks in Canada, Europe, and the Pacific
- The glue that holds the Internet together is the TCP/IP reference model and TCP/IP protocol stack
- To be on the Internet - a machine is on the Internet if it runs the TCP/IP protocol stack, has an IP address, and can send IP packets to all the other machines on the Internet.
- Up to 1990s, the Internet was largely populated by academic, government, and industrial researchers.
- WWW (World Wide Web) changed all that and brought millions of new, non-academic users to the net.
- ISP have POP (Point of Presence), where converted digital signals from the computer to analog signals (by modem) are removed from the telephone system and injected into the ISP's regional network, from this point on, the system is fully digital and packet switched
- The ISP's regional network consists of interconnected routers in the various cities the ISP serves.
- If the packet is destined for a host served directly by the ISP, the packet is delivered to the host. Otherwise, it is handed over to the ISP's backbone operator

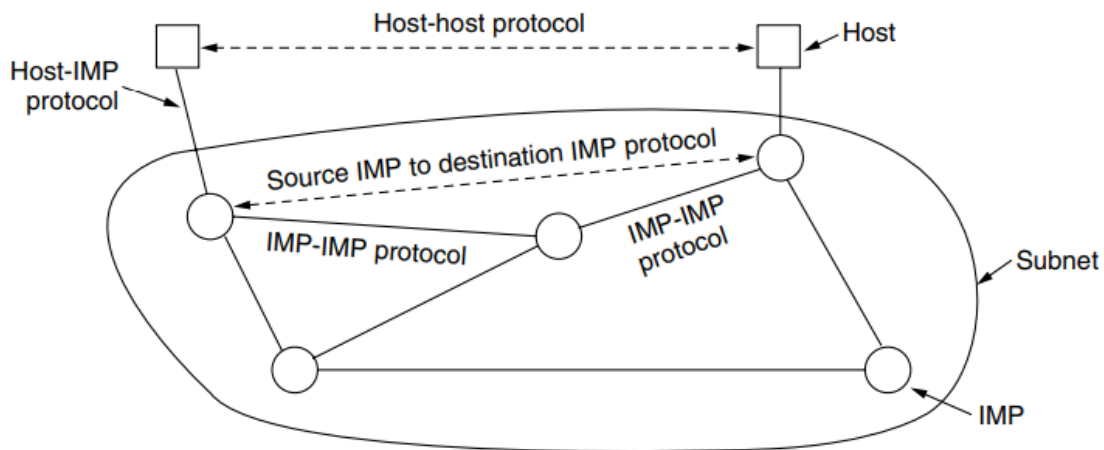
### The ARPANET

- The Internet is not a network, but a vast collection of different networks that use certain common protocols and provide certain common services.
- US created a single defence research organization, ARPA, the Advanced Research Projects Agency – it had no scientists or laboratories but did its work by issuing grants and contracts to universities and companies.



(a) Structure of the telephone system.

(b) Baran's proposed distributed switching system.



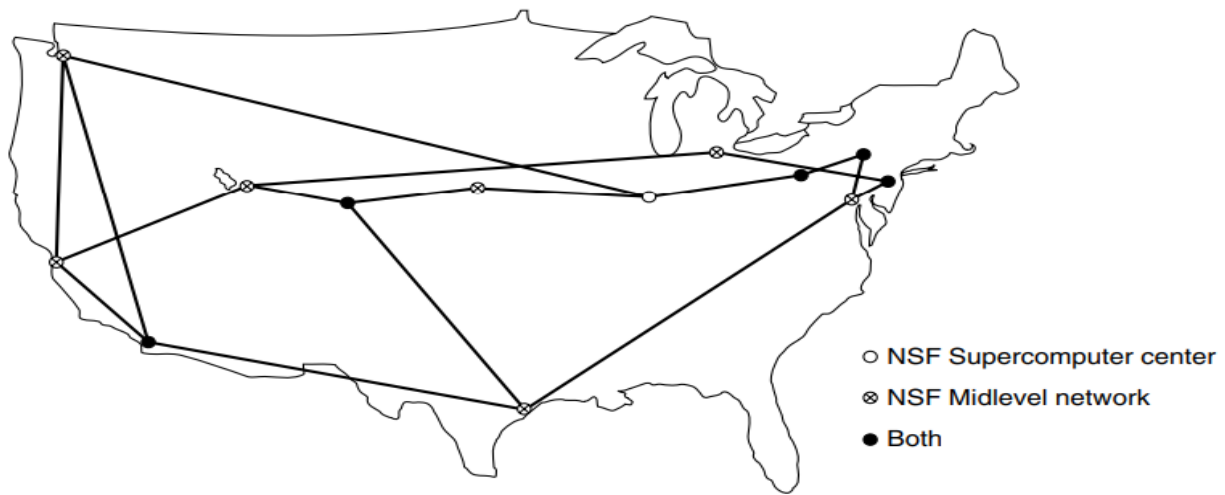
**Figure 1-26.** The original ARPANET design.

## DNS

- During the 1980s, additional networks, mainly LANs, were connected to the ARPANET. Finding hosts became increasingly expensive, so DNS (Domain Name System) was created to organize machines into domains and map host names onto IP addresses.
- DNS has become a generalized, distributed database system for storing a variety of information

## NSFNET

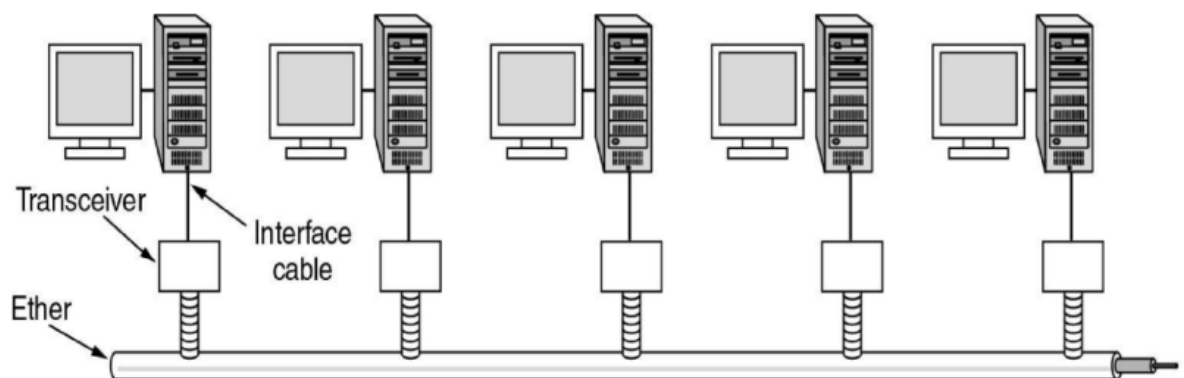
- In 1970s, NSF (the U.S. National Science Foundation) found the major impact of the ARPANET and designed a successor to the ARPANET that is open to all university research groups
- NSF decided to build a backbone network to connect its six supercomputer center, in San Diego, Boulder, Champaign, Pittsburgh, Ithaca, and Princeton
- NSF also funded some regional networks that connected to the backbone to allow users at thousands of universities, research labs, libraries, and museums to access any of the supercomputers and to communicate with one another.
- This complete network was called NSFNET
- NSF awarded contracts to four different network operators to establish a NAP (Network Access Point)
- During the 1990s, many other countries and regions also built national research networks, often patterned on the ARPANET and NSFNET. They are Europa NET and EBONE in Europe.



**Figure 1-28.** The NSFNET backbone in 1988.

## Ethernet

- Internet and ATM are designed for wide area networking
- The most popular LAN is Ethernet
- Up to 256 machines could be attached to the system via transceivers screwed onto the cable.
- A cable with multiple machines attached to it in parallel is called a multidrop cable
- A computer first listened to the cable to see if someone else was already transmitting, the computer held back until the current transmission finished
- Architecture of the original Ethernet.
- If two or more computers start transmitting at once each computer listens during its own transmission and if it detects interference, jam the ether to alert all senders
- Then the station/computer backs off and waits a random time before retrying
- If a second collision happens, the random waiting time is doubled, and so on, to spread out the competing transmissions and give one of them a chance to go first

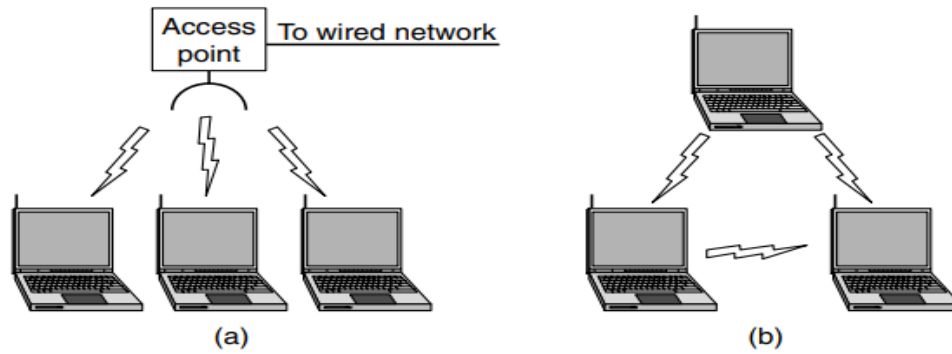


- In 1978 Xerox drew the 10-Mbps Ethernet standard – became IEEE 802.3 standard in 1983.

## Wireless LANs: 802.11

- IEEE committee that standardized the wired LANs was given the task of creating wireless LAN standard that resulted in 802.11
- Commonly known as WiFi
- The proposed standard had to work in two modes: In the presence of a base station In the absence of a base station

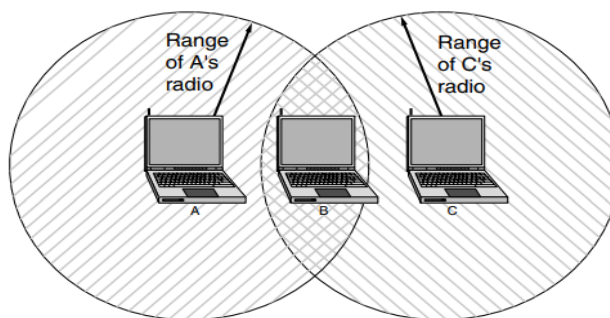
- In the first case, all communication go through the base station, called an access point
- In the second case, the computers would just send to one another directly - ad hoc networking.



(a) Wireless networking with a base station.

(b) Ad hoc networking.

- Ethernet had dominated local area networking, so the committee decided to make 802.11 compatible with Ethernet above the data link layer
- Possible to send an IP packet over the wireless LAN the same way a wired computer sent an IP packet over Ethernet .
- Another problem is that a radio signal can be reflected off solid objects, so it may be received multiple times (along multiple paths). This interference results in what is called multipath fading.



## 1.7 PHYSICAL LAYERS

### 1.7.1 THEORETICAL BASIS FOR DATA COMMUNICATION

- Information can be transmitted on wires by varying some physical property such as voltage or current.
- By representing the value of this voltage or current as a single-valued function of time,  $f(t)$ , we can model the behavior of the signal and analyze it mathematically.
- This analysis is the subject of the following sections.

#### **Fourier Analysis :**

In the early 19th century, the French mathematician Jean-Baptiste Fourier proved that any reasonably behaved periodic function,

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$

1.  $f$  is the function's fundamental frequency

2.  $T = 1/f$  is the function's period

3.  $a_n$  and  $b_n$  are the amplitudes of the  $n$ th harmonics

The series representation of  $g(t)$  is called its Fourier series expansion.

In communications, we can always represent a data signal using a Fourier series by imagining that the signal repeats the same pattern forever.

Moreover, we can compute the coefficients  $a_n$  and  $b_n$ :

$$a_n = \frac{2}{T} \int_0^T g(t) \sin(2\pi n f t) dt$$

$$b_n = \frac{2}{T} \int_0^T g(t) \cos(2\pi n f t) dt$$

$$c = \frac{2}{T} \int_0^T g(t) dt$$

### Bandwidth-Limited Signals

- The relevance of all of this to data communication is that real channels affect different frequency signals differently.
- Let us consider a specific example: the transmission of the ASCII character “b” encoded in an 8-bit byte. The bit pattern that is to be transmitted is 01100010.
- The left-hand part of Fig. 2-1(a) shows the voltage output by the transmitting computer. The Fourier analysis of this signal yields the coefficients:

$$a_n = \pi \cdot 1 [\cos(\pi n / 4) - \cos(3\pi n / 4) + \cos(6\pi n / 4) - \cos(7\pi n / 4)]$$

$$b_n = \pi \cdot 1 [\sin(3\pi n / 4) - \sin(\pi n / 4) + \sin(7\pi n / 4) - \sin(6\pi n / 4)]$$

$$c = 3/4$$

### Maximum data rate of a channel

- Every communication channel has a maximum data carrying capacity.
- Nyquist theorem states that for a noiseless channel,  
Maximum data rate =  $2B \log_2 V$  bits/sec.

Where,

$B$  = bandwidth

$V$  = number of signal levels

- This means increasing bandwidth or signal levels increases data rate.
- In practical systems, noise is always present, which affects transmission.
- Shannon's theorem gives the maximum data rate for a noisy channel:  
maximum number of bits /sec =  $B \log_2 (1 + S/N)$
- Here,  $S/N$  is the signal-to-noise ratio, usually expressed in decibels (dB).
- Hence, the data rate of a channel is limited by bandwidth and noise level.

### 1.7.2 GUIDED TRANSMISSION MEDIA

- The purpose of the physical layer is to transport bits from one machine to another.
- Various physical media can be used for the actual transmission.
- Each one has its own niche in terms of bandwidth, delay, cost, and ease of installation and maintenance.



- Media are roughly grouped into guided media, such as copper wire and fiber optics, and unguided media, such as terrestrial wireless, satellite, and lasers through the air.

## MAGNETIC MEDIA

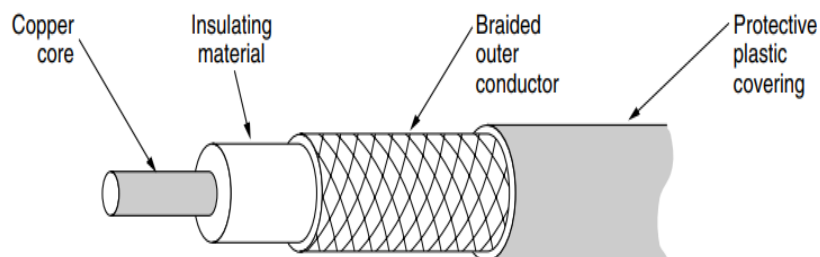
- One of the most common ways to transport data from one computer to another is to write them onto magnetic tape or removable media (e.g., recordable DVDs), physically transport the tape or disks to the destination machine, and read them back in again.
- Although this method is not as sophisticated as using a geosynchronous communication satellite, it is often more cost effective, especially for applications in which high bandwidth or cost per bit transported is the key factor.

## TWISTED PAIR

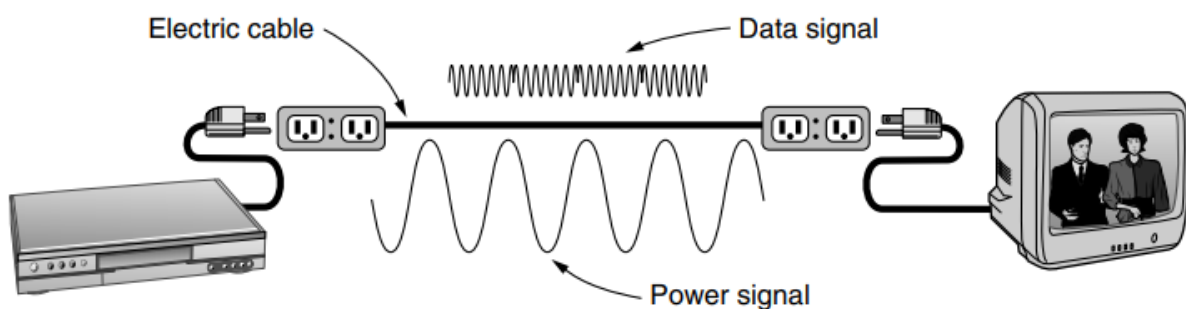
- One of the oldest and still most common transmission media is twisted pair.
- A twisted pair consists of two insulated copper wires, typically about 1 mm thick. The wires are twisted together in a helical form, just like a DNA molecule.
- The most common application of the twisted pair is the telephone system.
- Twisted pairs can be used for transmitting either analog or digital information.
- The bandwidth depends on the thickness of the wire and the distance traveled, but several megabits/sec can be achieved for a few kilometers in many cases.

## COAXIAL CABLE

- Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- The name of the cable is coaxial as it contains two conductors parallel to each other.
- It has a higher frequency as compared to Twisted pair cable.
- The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- The middle core is responsible for the data transferring whereas the copper mesh prevents from the EMI(Electromagnetic interference).



## FIBER OPTICS



- Fibre optic cable is a cable that uses electrical signals for communication.
- Fibre optic is a cable that holds the optical fibres coated in plastic that are used to send the data by pulses of light.
- The plastic coating protects the optical fibres from heat, cold, electromagnetic interference from other types of wiring.
- Fibre optics provides faster data transmission than copper wires.